

WORKSHEET 1: STATEMENTS, SETS, & GRAPHS

1. STATEMENTS

Throughout your lives, you will encounter many different notions of “truth.” In Mathematics however, we tend to be very literal.

DEFINITION. A mathematical **statement** is any sentence that has exactly one truth value. That is, we may *unambiguously*^a label that sentence with **T** or **F**.

^aUnambiguous means *not* open to more than one interpretation.

Exercise 1. *Determine whether or not the following sentences are mathematical statements. For each statement, determine the truth value.*

(1) $3^2 + 4^2 = 5^2$.

(2) *Why are you reading this?*

(3) *The San Francisco Giants are the best baseball team.*

(4) $(2 + 3)^2 = 2^2 + 3^2$.

(5) *Sara (your instructor) is six feet tall.*

(6) $|x - 3| \leq 100$.

(7) *There are infinitely many prime numbers.*

(8) *817326745 is an even number.*

(9) *n is an odd number.*

(10) $12!$

(11) *Spiderman.*

(12) *It's always sunny in Philadelphia.*

(13) *Every prime number is odd.*

(14) *Isaiah (your instructor) is less than six feet tall.*

(15) *This sentence is false.*¹

¹Once you've pondered this for a few minutes, read the first paragraph of the Wikipedia page titled “paradox.”

2. CONDITIONAL STATEMENTS AND PROOF

One of the most important types of statements in mathematics are called **conditional statements**. These are statements of the form *if P then Q* , which we denote by $P \Rightarrow Q$ (sometimes read as P implies Q). In this form, the statement P is called the **hypothesis** and statement Q is known as the **conclusion**.

Example. *The following statements are conditional.*

- (1) *If you are in Nebraska, then you are in the United States.*
- (2) *If $x > 0$, then $x + 1 > 1$.*
- (3) *n^2 is odd, provided that n is odd.*
- (4) *A division ring is a field only if the division ring is finite.*
- (5) *Whenever it rains, Sara brings her umbrella.*
- (6) *Isaiah wears a hat, provided he doesn't want to comb his hair.*
- (7) *For an integer x , being negative is a sufficient condition for x^3 to be negative.*
- (8) *A nonzero determinant is necessary for a matrix to be invertible.*

Exercise 2. *Identify the hypotheses and conclusions of the above statements (it may be helpful to rephrase the sentences in the form “if P then Q ,” without changing the meaning). Write three of your own conditional statements. Are the statements true?*

In mathematics, we often seek to *prove* such conditional statements. That is, given a hypothesis which we assume to be true, we seek to develop an argument that demonstrates that our desired conclusion must also be true. That is, we show $P \Rightarrow Q$ is true. When we discuss negation tomorrow, we will see that one can equivalently show that not Q implies not P .

Exercise 3. *Determine whether the following conditional statements are true or false. Justify.*

- (1) *If a graph has a vertex of degree 1, then there is another vertex of at least degree 1.*
- (2) *If a graph has a vertex of degree 0, it is disconnected.*
- (3) *If a graph with 5 vertices has a vertex of degree 4, then the graph is complete.*

(4) If a graph has a vertex of degree 1, then it has a bridge.

(5) If $G = K_n$ is the complete graph on n many vertices, then the number of edges in the graph G is $\frac{n(n-1)}{2}$.²

(6) A connected graph in which every edge is a bridge is called a tree. Determine if the following conditional statement is true: If T is a tree with n vertices, then T has $n - 1$ edges.

DEFINITION. An integer z is called **even** if there exists an integer k such that $z = 2k$.

Exercise 4. Using the definition, justify why 6 is even.

Exercise 5. Prove that the sum of two even numbers is even. Prove that the product of two even numbers is even. Is it true that if z is even, then so is $z/2$?

Exercise 6. Prove that if z is even, then z^n is even, for any positive integer $n \geq 1$.³

DEFINITION. An integer z is called **odd** if there exists an integer k such that $z = 2k + 1$.

Exercise 7. Prove that the product of an even number and an odd number is even.

Exercise 8. Prove that the product of two odd numbers is odd.

Exercise 9. Prove that the sum of an odd number and an even number is still odd.

Exercise 10. Suppose z is odd.

(a) Prove that z^2 is odd.

(b) Prove that z^3 is odd.

(c) How can you prove that z^n is odd for all integers $n \geq 1$?

²*Hint:* Think about small values of n first.

³*Hint:* The case $n = 1$ is easy. See if you can prove this for $n = 2$ first. How can you make your proof work for all integers $n \geq 1$?